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Carbon Emission Tracking and Environmental Compliance System for Industries

DSA0104 - Object Oriented Programming With C++

Guided By:
Dr. Devan

Presented by:
Harshavardhan R V (192472163)
Nawab Saad (192472266)
Pavan Jana (192472313)

INTRODUCTION

- Industries use fuel and electricity for their daily operations.
- These activities produce carbon dioxide (CO₂) emissions.
- Monitoring carbon emissions is important for protecting the environment.
- Manual calculation and record keeping can be difficult.
- Our project develops a C++ system to track industrial emissions and check environmental compliance.

OBJECTIVES

- To collect industrial activity data.
- To store and manage emission-related data.
- To calculate CO₂ emissions.
- To track emissions over time.
- To compare emissions with environmental limits.

PROBLEM STATEMENT

- Industries generate carbon emissions from different activities.
- Emission data may be stored manually or in separate files.
- Calculating total emissions manually can cause errors.
- It is difficult to continuously track emission levels.
- Industries need to know whether their emissions are within permitted/reference limits.

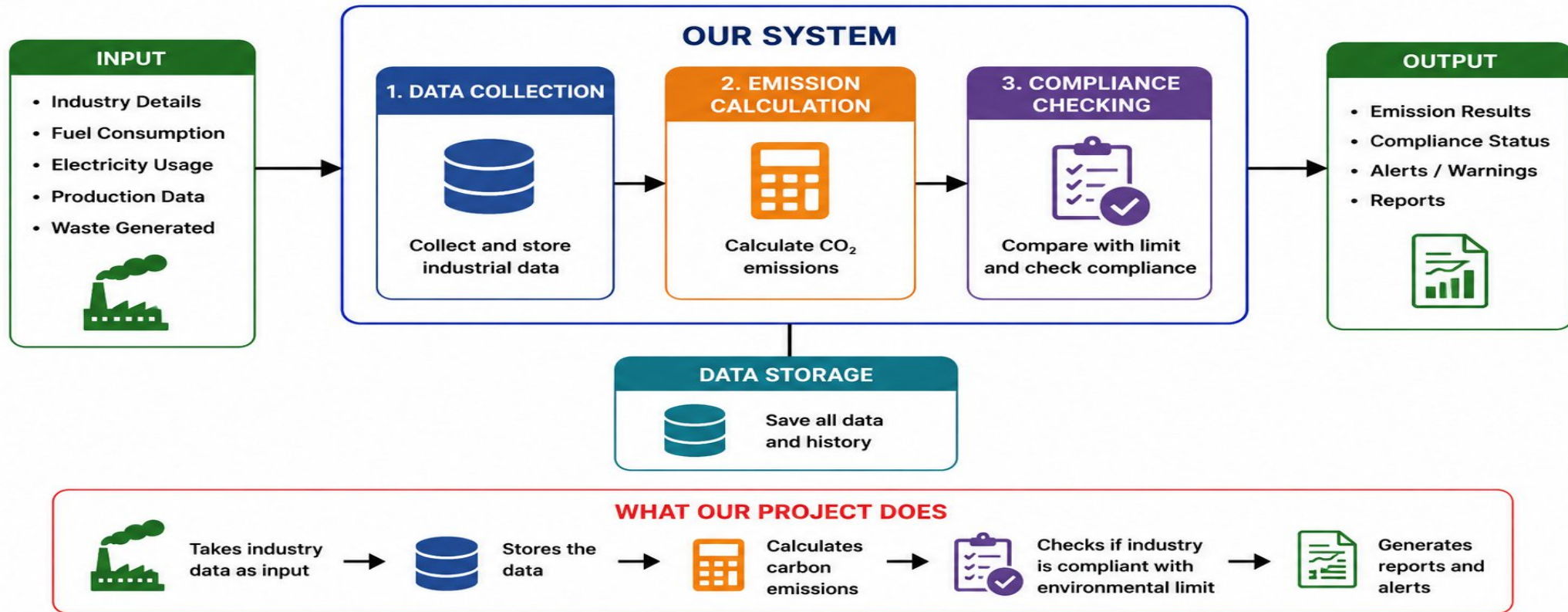
Literature Review

S. No	Author(s) & Year	Focus Area	Methods Used	Key Findings	Research Gap
1	Adhikari et al., 2025	Carbon Footprint Tracking in Industries	IoT sensors, AI analytics, cloud monitoring	Real-time monitoring improves emission measurement and sustainability.	Limited integration with regulatory compliance and automated reporting.
2	Striletskyi, 2024	Enterprise Greenhouse Gas Emission Management	Emission monitoring algorithms, data analytics	Automated systems improve greenhouse gas tracking accuracy.	Lacks predictive analytics and intelligent decision support.
3	Cenci & Biffis, 2025	Greenhouse Gas Reporting Standards	Emission reporting framework, comparative analysis	Standardized reporting enhances environmental transparency and compliance.	Limited support for continuous real-time industrial monitoring.
4	Alzoubi et al., 2026	Carbon Footprint Monitoring and Privacy	Carbon monitoring tools, privacy risk assessment	Secure monitoring systems improve trust in environmental data.	Insufficient focus on industrial compliance automation and alerts.
5	IEEE Standards Association, 2025	Blockchain-Based Carbon Emission Management	Blockchain, distributed ledger, secure data sharing	Blockchain improves transparency and tamper-proof emission records.	Limited deployment in large-scale industrial environmental monitoring.

System Architecture

CARBON EMISSION TRACKING AND ENVIRONMENTAL COMPLIANCE SYSTEM FOR INDUSTRIES

— SYSTEM ARCHITECTURE —



LIST OF MODULES

1. Industrial Data Management Module
2. Carbon Emission Calculation & Tracking
3. Environmental Compliance and Reporting Module

MODULES 1: Industrial Data Management

- Collects industry details.
- Records fuel consumption.
- Records electricity consumption.
- Records production/activity data.
- Stores environmental data.
- Allows searching and updating records.



Module 2: Carbon Emission Calculation & Tracking

- Takes activity data from Module 1.
- Applies emission factors.
- Calculates CO₂ emissions.
- Calculates total emissions.
- Tracks monthly and yearly emissions.
- Identifies high-emission activities.



Module 3: Environmental Compliance & Reporting

- Takes calculated emissions.
- Compares emissions with a reference/environmental limit.
- Determines compliance status.
- Identifies excess emissions.
- Generates warnings.
- Generates reports.



OUTPUT

MODULE 01

Industrial Dataset Input

Upload an industrial dataset in CSV format. The website automatically calculates CO₂ emissions, stores the records, and updates the compliance dashboard.



Upload Dataset

Select a **CSV dataset**. Required columns: **Industry Name, Industry Type, Fuel Consumption, Electricity Consumption, Production, Month**.

Choose File No file chosen



Expected Dataset Format

Your CSV should contain one industrial record per row.

```
Industry Name,Industry Type,Fuel Consumption,Electricity  
Consumption,Production,Month  
ABC Manufacturing,Manufacturing,1200,5000,500,2026-08  
XYZ Cement,Cement,900,4200,700,2026-08
```

 Load Sample Dataset

 Download CSV Template

 Clear Saved Dataset

✓ Sample dataset loaded and automatically processed.

MODULE 02

Carbon Emission Calculation & Tracking

CO₂ is calculated automatically for every dataset record using the configured emission factors.

Electricity Factor

0.82

kg CO₂ / kWh

Fuel Factor

2.31

kg CO₂ / litre

Dataset Emission

85,061.5

kg CO₂ total

Formula used:

CO₂ = (Electricity × 0.82) + (Fuel × 2.31)

Prototype factors for demonstration; replace with approved factors for your implementation.

Environmental Compliance & Reporting

Check reference limits and manage environmental records.

Saved Industrial Records

 Download Report

 Clear All

ID	Industry	Type	Fuel	Electricity	Production	CO ₂	Status	Month	Action
1788362517760	Power Unit A	Power Plant	2500 L	9000 kWh	1100 T	13155 kg	 EXCEEDED	2026-07	Delete
1788362517759	Sun Textile	Textile	400 L	2500 kWh	300 T	2974 kg	 EXCEEDED	2026-07	Delete
1788362517758	Green Steel Ltd	Steel	1800 L	7000 kWh	900 T	9898 kg	 EXCEEDED	2026-08	Delete
1788362517757	XYZ Cement	Cement	900 L	4200 kWh	700 T	5523 kg	 EXCEEDED	2026-08	Delete
1788362517756	ABC Manufacturing	Manufacturing	1200 L	5000 kWh	500 T	6872 kg	 EXCEEDED	2026-08	Delete

FUTURE SCOPE

- Add advanced emission prediction using historical data.
- Support multiple industrial emission factors and standards.
- Develop graphical dashboards for emission visualization.
- Add database integration for large-scale data storage.
- Extend the system for automated compliance monitoring

CONCLUSION

- Our project provides a C++-based system for tracking industrial carbon emissions.
- It collects fuel, electricity, and other activity data from industries.
- The system calculates CO₂ emissions using emission factors and stores the results.
- Monthly or yearly records can be compared track whether emissions increase or decrease.
- Finally, it generates emission reports, compliance status.

REFERENCES

1. N. Adhikari, H. Li, and B. Gopalakrishnan, “A Bibliometric and Systematic Review of Carbon Footprint Tracking in Cross-Sector Industries: Emerging Tools and Technologies,” *Sustainability*, vol. 17, no. 9, Art. no. 4205, May 2025.
2. M. Striletskyi, “Algorithm for Enterprise Greenhouse Gas Emissions Control and Management System,” in *Proc. IEEE ACIT 2024*, 2024, pp. 631–635, doi: 10.1109/ACIT62333.2024.10712452.
3. Y. I. Alzoubi, A. E. Topcu, and E. Elbaşı, “From Measurement to Monitoring: Evaluating Privacy Risks in Carbon Footprint Tools,” *IEEE Access*, vol. PP, no. 99, pp. 1–1, Jan. 2026.
4. S. Cenci and E. Biffis, “Lack of Harmonisation of Greenhouse Gases Reporting Standards and the Methane Emissions Gap,” *Nature Communications*, vol. 16, Art. no. 1537, Feb. 2025.
5. IEEE Computer Society, *IEEE 3241.01-2024: IEEE Standard for Using Blockchain in Low Carbon Zones Evaluation*, IEEE Standards Association, 2025.

A soft, painterly splash of teal and light blue watercolor paint, centered on a white background. The paint has a textured, organic feel with varying shades of blue and green.

Thank You